INTERACTIVE GAME ARCHITECTURE SPECIFICATION

NOVOSRITI — MISSION LAB

1. PROJECT OVERVIEW

1.1 Mission Selection Screen

The Mission Lab begins with a mission selection screen where users can choose from five historical space exploration missions. The design is simple and attractive, specifically tailored for school-age users.

#	Mission Name	Location	Card Elements Required	Environment Visuals
1	Apollo Laser Reflectors	Moon	• Mission Name • Spacecraft / Instrument Image • Location Tag (Moon / Mars) • One-Line Mission Objective • Start Mission Button	**Moon:** Grey, white, and black surface; no visible atmosphere.
2	Apollo Seismometers	Moon		
3	Viking Landers	Mars		**Mars:** Red/orange landscape, dusty atmosphere, and rocky terrain.
4	Spirit & Opportunity	Mars		
5	Ingenuity (Helicopter Drone)	Mars		

1.2 Common Player Journey

Every mission follows a standard five-stage structure while maintaining unique mission-specific objectives and interaction mechanics:

1. **Mission Briefing:** Explain the real-world historical mission.
2. **Landing / Deployment:** Introduce the spacecraft or specialized equipment.
3. **Exploration:** Allow the player to interact directly with the environment.
4. **Main Objective:** Complete the designated scientific task.
5. **Mission Complete:** Present final results and educational key takeaways.

2. COMMON GAMEPLAY SYSTEM

2.1 Shared Controls

A consistent control scheme is used wherever movement apply. *Note: Viking landers and Apollo instrument deployments utilize dedicated interaction interfaces rather than spatial movement controls.*

Keyboard / Mouse Input	Mobile Input	Action
W / Up Arrow (↑)	On-Screen D-Pad / Stick	Move Forward
S / Down Arrow (↓)	On-Screen D-Pad / Stick	Move Backward
A / Left Arrow (←)	On-Screen D-Pad / Stick	Move Left
D / Right Arrow (→)	On-Screen D-Pad / Stick	Move Right
E	On-Screen Context Button	Interact
Mouse Click	Touch Selection	Select Objects / Activate Tools
Space	On-Screen Action Button	Special Action (if required)

2.2 Contextual Action Rule

Interface action buttons are hidden by default to keep the screen uncluttered. Action controls appear dynamically only when the player is within range of an interactable object or area:

- **Near a rock:** Displays *Inspect Rock*
- **At reflector deployment point:** Displays *Place Reflector*
- **Near soil collection point:** Displays *Collect Soil*
- **Near sample tube:** Displays *Store Sample*
- **At helicopter launch area:** Displays *Start Flight*

2.3 Target and Objective System

Each mission displays the current objective, overall mission progress, target markers, and completion status. For example, the objective sequence for *Spirit & Opportunity* progresses as: **1. Reach target → 2. Capture photograph → 3. Inspect rock → 4. Complete investigation.**

2.4 Arrival Sequences

Short introductory animations establish the setting upon starting a mission:

- **Apollo:** Astronaut arrives on the Moon.
- **Viking:** Lander touches down on Mars.
- **Spirit & Opportunity / Curiosity / Perseverance:** Rover begins surface exploration.
- **Ingenuity:** Helicopter deploys from Perseverance.

2.5 Accessibility and Performance Requirements

- Use readable typography, large action buttons, and simple pre-mission instructions.
- Ensure low-end computer playability via lightweight animations and optimized physics.
- Full support for desktop (keyboard/mouse) and mobile (touch buttons) input.

3. APOLLO LASER REFLECTORS

Mission Title: PLACE THE MIRROR THAT SENDS LIGHT HOME

> **Primary Objective:** Deploy a lunar retroreflector array and complete a laser-ranging demonstration.
> **Setting:** Moon surface.
> **Gameplay Type:** Astronaut movement, object placement, and laser-return interaction.

3.1 Real Mission Context & Scientific Accuracy

Apollo astronauts deployed passive laser retroreflector arrays on the Moon containing corner-cube reflectors. Observatories on Earth emit laser pulses towards these units to calculate Earth–Moon distance and lunar orbital motion based on round-trip light timing.

> **Important Scientific Accuracy Guidelines:**
> - The laser source is situated on **Earth**, not the Moon.
> - The reflector is deployed on the **Moon** to bounce incoming light back towards Earth.
> - The reflector array is entirely passive (requires no battery or computer).
> - The Moon surface environment does not generate the laser pulse.

3.2 REQUIRED GAMEPLAY SEQUENCE

PHASE / TRIGGER	PLAYER SEES	PLAYER ACTION	GAME RESPONSE / NEXT STATE
1. Mission briefing	Mission purpose, a simple reflector diagram, and two objectives.	Press “Start Mission”.	Begin the Moon arrival cinematic.
2. Lunar landing	Lunar Module descends and touches down. A “Landing Complete” message appears.	No control required; allow “Skip”.	Show the astronaut at the lander hatch with an “Exit Lander” button.
3. Exit lander	Astronaut at the hatch and a highlighted deployment location on the ground.	Press “Exit Lander”, then use directional controls to walk to the highlighted location.	Astronaut steps onto the surface. When inside the highlighted location, show “Checkpoint Received” and mark the Lunar Module as the next destination.
4. Return to lunar module	Checkpoint confirmation and a marker at the Lunar Module equipment bay.	Use directional controls to walk back to the Lunar Module.	When inside the equipment-bay marker, show “Pick Up Reflector”.
5. Pick up equipment	Reflector package stored on the Lunar Module.	Press “Pick Up Reflector”.	Reflector package becomes attached to or carried by the astronaut. Update objective 1 to Complete; show “Move to Deployment Point”.
6. Deploy reflector	A placement outline appears on the ground.	Move into the marked zone and press “Place Reflector”.	Play a short placement animation; objective 2 becomes Complete.
7. Laser demonstration	A small Earth / observatory inset and a “Send Laser Pulse” button.	Press “Send Laser Pulse”.	Animate a pulse traveling from Earth to the reflector and returning. Show a timer / round-trip concept.
8. Learning result	A result card explains that round-trip light time is used to calculate distance.	Press “Continue”.	Open the mission completion screen.

3.3 BUTTON AND STATE RULES

- **Start Mission** is visible on the briefing screen only.
- **Exit Lander** appears only after the “Landing Complete” message.
- **Checkpoint Received** is shown when the astronaut first enters the highlighted deployment location.
- **Pick Up Reflector** appears only at the Lunar Module, after the checkpoint is received.
- **Place Reflector** appears only inside the deployment zone while carrying the package.
- **Send Laser Pulse** appears only after the reflector has been placed.
- The laser originates from the Earth / observatory inset, not from the astronaut or the lunar panel.

3.4 REQUIRED OBJECTIVES

OBJECTIVE	COMPLETION CONDITION
01 - Reach the deployment area	Astronaut enters the marked area with the reflector package.
02 - Place the reflector array	Player confirms placement inside the target zone.
03 - Return a laser pulse	Player triggers the Earth-based laser demonstration and watches the return signal.

3.5 COMPLETION SCREEN

Title: MISSION COMPLETED - LUNAR LASER REFLECTOR DEPLOYED
Learning card: “The reflector has no power supply. Its corner cubes return laser light toward the source. Scientists use the travel time of the light to measure the Earth-Moon distance.”
Buttons: “Replay Mission” and “Return to Mission Selection”.

ACCURACY NOTE
Do not show the Moon panel generating a laser. It only reflects a pulse sent from Earth. Do not claim that the game has performed a real scientific measurement. The animation is a simplified demonstration.

3.6 OPTIONAL ENHANCEMENT

Show a simple timer that counts the pulse travel time. If an approximate value is shown, label it as an educational illustration and verify the value before publication.

4. APOLLO SEISMOMETERS

Mission title: LISTEN TO THE MOON

FIELD	SPECIFICATION
Primary objective	Deploy a passive seismic experiment and interpret a lunar vibration signal.
Mission setting	Moon surface. Use an Apollo 12-style ALSEP station as the main playable example. Apollo 11's early seismic package was simpler and should not be represented as the full later ALSEP station.
Gameplay type	Astronaut movement, experiment deployment, station activation, and waveform interpretation.

4.1 REAL MISSION CONTEXT

Apollo lunar seismometers detected ground vibrations, including moonquakes and impacts. The measurements helped scientists study how seismic waves traveled through the Moon and learn about its internal structure. Later Apollo missions deployed more advanced seismic stations as part of the Apollo Lunar Surface Experiments Package (ALSEP).

4.2 REQUIRED GAMEPLAY SEQUENCE

PHASE / TRIGGER	PLAYER SEES	PLAYER ACTION	GAME RESPONSE / NEXT STATE
1. Mission briefing	A station diagram and the objective: deploy the listening instrument.	Press "Start Mission".	Begin lunar arrival and landing.
2. Lunar landing	Lunar Module on the surface; deployment equipment visible.	Skip cinematic or continue.	Show the astronaut at the lander hatch with an "Exit Lander" button.
3. Exit lander	Astronaut at the hatch and a highlighted station site a short distance from the lander.	Press "Exit Lander", then use directional controls to walk to the highlighted site.	Astronaut steps onto the surface. When inside the highlighted site, show "Checkpoint Received" and mark the Lunar Module as the next destination.
4. Return to lunar module	Checkpoint confirmation and a marker at the Lunar Module equipment bay.	Use directional controls to walk back to the Lunar Module.	When inside the equipment-bay marker, show "Collect Equipment".
5. Collect equipment	Seismometer and station equipment stored on the Lunar Module.	Press "Collect Equipment".	Equipment is carried by the astronaut. Show "Move to Station Site".
6. Move to station site	The highlighted station site, a short distance from the lander.	Move using directional controls.	At the marker, show "Deploy Seismometer".
7. Deploy seismometer	A placement outline and a simplified sensor model.	Press "Deploy Seismometer".	Play deployment animation. Update objective 1.
8. Connect the station	Cable / central station / antenna interaction points.	Press "Connect to Central Station", then "Deploy Antenna".	Show a status sequence: Sensor Connected / Station Ready.
9. Start listening	A "Start Listening" button and a quiet status panel.	Press "Start Listening".	Show a short listening interval, then reveal a seismic signal event.
10. Read the waveform	Waveform graph with a highlighted signal and simple prompt.	Choose "Signal Detected" or press "Continue" after viewing the trace.	Display a brief explanation of what the sensor measured.
11. Learning result	A Moon interior learning card.	Press "Continue".	Open completion screen.

4.3 BUTTON AND STATE RULES

- **Exit Lander** appears only after the lunar landing is complete.
- **Checkpoint Received** is shown when the astronaut first enters the highlighted station site.
- **Collect Equipment** appears only at the Lunar Module, after the checkpoint is received.
- **Deploy Seismometer** appears only at the marked location while carrying the equipment.
- **Connect to Central Station** and **Deploy Antenna** appear after the sensor is placed.
- **Start Listening** appears only when all station setup steps are complete.
- The waveform should be a real archived trace if the team can source and label one. Otherwise, label it clearly: "Illustrative waveform - not a live Apollo recording."

4.4 REQUIRED OBJECTIVES

OBJECTIVE	COMPLETION CONDITION
01 - Deploy the seismometer	Player places the instrument in the marked zone.
02 - Prepare the station	Player completes the central-station / antenna setup interaction.
03 - Detect a vibration signal	Player starts listening and views the waveform result.

4.5 COMPLETION SCREEN

Title: MISSION COMPLETED - LUNAR SIGNAL DETECTED
Learning card: “A seismometer measures ground vibrations. Apollo instruments detected moonquakes and other events. Scientists used the wave patterns to investigate the Moon’s interior.”
Buttons: “Replay Mission” and “Return to Mission Selection”.

ACCURACY NOTE
Do not let the player create a real moonquake. The player deploys and listens to a sensor. A generated waveform must be marked as illustrative. Use an Apollo 12-style ALSEP station for the main scenario; do not mix the simpler Apollo 11 package with later station hardware.

5. VIKING LANDERS

Mission title: SEARCH FOR SIGNS OF LIFE

FIELD	SPECIFICATION
Primary objective	Land a stationary spacecraft, collect Martian soil, and run a simplified biological / chemical experiment.
Mission setting	Mars surface. The two Viking landers arrived at different locations. For a short prototype, use one representative landing site and identify it as Viking 1 or Viking 2.
Gameplay type	Landing sequence, stationary lander interaction, robotic sampler arm, and sample-result interpretation.

5.1 REAL MISSION CONTEXT

Viking 1 and Viking 2 were lander-and-orbiter missions. The landers photographed the surface, studied the atmosphere and weather, scooped soil with a robotic arm, and carried experiments designed to look for signs of life. The experiments produced interesting chemical results but no clear evidence of living microorganisms at the tested sites.

5.2 REQUIRED GAMEPLAY SEQUENCE

PHASE / TRIGGER	PLAYER SEES	PLAYER ACTION	GAME RESPONSE / NEXT STATE
1. Mission briefing	Viking lander image, mission purpose, and three objectives.	Press “Start Mission”.	Begin launch / cruise transition.
2. Mars approach	Mars grows in view; a landing-zone selection overlay appears.	Choose one of two safe landing zones, or press “Continue” if using a fixed site.	Play entry, parachute, and powered-descent animation. Treat steering as a simplified game choice, not a true reconstruction.
3. Touchdown	Lander settles on the surface. A “Lander Ready” status appears.	Press “Begin Experiments”.	Camera and sampler controls become available. The lander remains stationary.
4. Photograph the site	A “Take Panorama” button and camera target.	Press “Take Panorama”.	Show a prepared surface image or simulated camera frame. Mark objective 1 complete.
5. Choose soil location	Two or three sample patches appear on a small nearby map / camera view.	Select one sample patch.	Move the robotic arm toward the chosen patch.
6. Scoop soil	Arm position and a “Collect Soil” prompt.	Press “Collect Soil”.	Play scoop animation and move soil into the lander's instrument inlet. Mark objective 2 complete.
7. Run experiment	A simple instrument panel with “Start Analysis”.	Press “Start Analysis”.	Show a short analysis sequence and a result card. Mark objective 3 complete.
8. Send results	A “Transmit Results” button.	Press “Transmit Results”.	Show a radio / data transmission animation and open completion screen.

5.3 BUTTON AND STATE RULES

- After touchdown, do not show rover driving controls. Viking was a stationary lander.
- **Take Panorama** appears after the lander reports ready.
- **Collect Soil** appears after the player selects a reachable sample patch and the arm is positioned.
- **Start Analysis** appears after soil has been delivered to the instrument inlet.
- **Transmit Results** appears after the experiment result is displayed.

5.4 REQUIRED OBJECTIVES

OBJECTIVE	COMPLETION CONDITION
01 - Photograph the landing site	Player captures the site panorama.

OBJECTIVE	COMPLETION CONDITION
02 - Collect a soil sample	The robotic arm scoops and delivers soil to the inlet.
03 - Analyze the sample	Player runs the experiment and views the result.

5.5 COMPLETION SCREEN

Title: MISSION COMPLETED - SOIL EXPERIMENT COMPLETE
Learning card: “Viking tested Martian soil for possible signs of life. The results were scientifically interesting, but they did not provide clear evidence of living microorganisms at the tested sites.”
Buttons: “Replay Mission” and “Return to Mission Selection”.

ACCURACY NOTE
Do not write “Viking proved there is no life on Mars.” The experiments did not provide clear evidence of life at the tested locations. Viking is a lander, not a mobile rover; keep it stationary after touchdown.

6. SPIRIT & OPPORTUNITY

Mission title: EXPLORE THE RED PLANET

FIELD	SPECIFICATION
Primary objective	Drive across the Martian surface, photograph selected targets, and inspect a rock surface.
Mission setting	Mars surface. Let the player choose Spirit (Gusev Crater) or Opportunity (Meridiani Planum) before starting. The same basic rover control system may be reused for both.
Gameplay type	Rover driving, camera capture, rock inspection, and a short Rock Abrasion Tool interaction.

6.1 REAL MISSION CONTEXT

Spirit and Opportunity were twin Mars Exploration Rovers. They studied Martian rocks, soil, and landscapes to investigate the planet's geological history and evidence of past water. Their Pancam cameras captured panoramic images. Their Rock Abrasion Tool (RAT) ground away a small amount of rock surface to expose fresher material for inspection; it was not a sample-return drill.

6.2 ROVER CHOICE SCREEN

- Show two selectable cards: Spirit and Opportunity.
- Spirit card: location label “Gusev Crater”; objective hint “Explore rocks and clues of past water.”
- Opportunity card: location label “Meridiani Planum”; objective hint “Explore layered rocks and minerals linked to past water.”
- The selected rover starts the same control system with a different mission location and target descriptions.

6.3 REQUIRED GAMEPLAY SEQUENCE

PHASE / TRIGGER	PLAYER SEES	PLAYER ACTION	GAME RESPONSE / NEXT STATE
1. Mission briefing	Spirit / Opportunity choice and short science objective.	Select one rover and press “Start Mission”.	Begin Mars launch / entry / landing cinematic.
2. Touchdown	Rover on the Martian surface and a mini-map with two target pins.	Press “Begin Exploration”.	Enable driving controls.
3. Drive to target A	Directional controls and target A marker.	Drive forward / backward / turn until inside the target zone.	Show “Take Photo” when the rover reaches the zone.
4. Capture target A	Camera framing prompt.	Press “Take Photo”.	Display a prepared mission-style surface image. Mark objective 1 complete.
5. Drive to target B	Target B pin becomes active.	Drive to the second target zone.	Show “Take Photo” and “Inspect Rock” when close enough.
6. Capture target B	Camera framing prompt.	Press “Take Photo”.	Display second image. Mark objective 2 complete.
7. Inspect rock (RAT)	A close rock target and “Use Rock Abrasion Tool” button.	Press the tool button, then “View Fresh Surface”.	Play a brief grinding / brushing-style animation and show a close-up view. Do not animate a deep core sample.
8. Learning result	A short card about rock surfaces and past water clues.	Press “Continue”.	Open completion screen.

6.4 BUTTON AND STATE RULES

- Use the same movement controls for Spirit and Opportunity; swap the map, target labels, and relevant mission facts.
- **Take Photo** appears only when the rover enters a photo target zone.
- **Inspect Rock** or **Use Rock Abrasion Tool** appears only near the marked rock.

• Do not add a standard soil-scoop objective. The RAT grinds the surface of a rock; it does not collect a sealed core sample for return to Earth.

6.5 REQUIRED OBJECTIVES

OBJECTIVE	COMPLETION CONDITION
01 - Reach and photograph target A	Player drives into target A zone and captures an image.
02 - Reach and photograph target B	Player drives into target B zone and captures an image.
03 - Inspect a rock surface	Player activates the RAT interaction and views the exposed surface.

6.6 COMPLETION SCREEN

Title: MISSION COMPLETED - MARTIAN TERRAIN EXPLORED
Learning card: “Spirit and Opportunity used cameras and scientific tools to study Martian rocks and soil. The RAT removed dusty, weathered rock surfaces so instruments could inspect fresher material.”
Buttons: “Replay Mission” and “Return to Mission Selection”.

ACCURACY NOTE
The rover may take a photo in the game, but show it as a captured / selected camera view using a credited image asset or a clearly labeled simulation. Do not imply a real-time NASA camera feed. The RAT is a rock-grinding tool, not a deep drilling and sample-caching system.

7. INGENUITY HELICOPTER

Mission title: THE FIRST POWERED FLIGHT ON ANOTHER WORLD

FIELD	SPECIFICATION
Primary objective	Deploy the helicopter from Perseverance, complete a short flight route, capture an aerial view, and land safely.
Mission setting	Mars surface and low-altitude flight above Jezero Crater. Ingenuity arrived attached beneath Perseverance and was later deployed on the surface.
Gameplay type	Deployment, rotor spin-up, controlled takeoff, simple flight navigation, aerial photography, and landing.

7.1 REAL MISSION CONTEXT

Ingenuity was a technology demonstration helicopter carried to Mars by the Perseverance rover. It demonstrated powered, controlled flight in the atmosphere of another planet. NASA originally planned up to five flights; Ingenuity completed 72 flights. Its final flight was on January 18, 2024, and rotor-blade damage ended its ability to fly. The mission was declared complete on January 25, 2024.

7.2 REQUIRED GAMEPLAY SEQUENCE

PHASE / TRIGGER	PLAYER SEES	PLAYER ACTION	GAME RESPONSE / NEXT STATE
1. Mission briefing	Ingenuity / Perseverance connection, the objective: test flight, and three objectives.	Press “Start Mission”.	Begin launch / cruise transition. Ingenuity travels stowed beneath Perseverance.
2. Mars approach	Mars grows in view; a landing-zone selection overlay appears for Perseverance, with Ingenuity stowed beneath it.	Choose one of two safe landing zones, or press “Continue” if using a fixed site.	Play entry, parachute, and powered-descent animation. Treat steering as a simplified game choice, not a true reconstruction. On touchdown, show Ingenuity stowed beneath Perseverance on Mars.
3. Deployment	A release / deployment prompt after Perseverance is stationary.	Press “Deploy Ingenuity”.	Animate the helicopter being lowered / placed onto the surface. Do not show a separate Earth launch for Ingenuity.
4. Pre-flight check	Rotor, battery, and status panel.	Press “Run Pre-flight Check”.	Show simplified green status indicators. These are game indicators, not real mission telemetry.
5. Rotor spin-up	A “Spin Rotors” button and rotor animation.	Press “Spin Rotors”.	Animate rotor blades spinning. Once ready, show “Take Off”.
6. Take off	Takeoff button and a marked flight height / waypoint.	Press “Take Off”, then use up / down controls to reach the safe height.	Switch to flight controls and show the route indicator.
7. Fly to waypoint	Two or three aerial waypoints.	Use directional controls to move through each waypoint.	Mark each waypoint complete. If the helicopter drifts, allow correction without instant failure.
8. Capture aerial image	A “Capture Image” prompt at the final waypoint.	Press “Capture Image”.	Show a prepared aerial view / simulated camera frame and mark objective 2 complete.
9. Land safely	A marked landing zone and “Land” button.	Move above the landing zone and press “Land”.	Animate descent and touchdown. Mark objective 3 complete.
10. Legacy result	A short historical achievement card.	Press “Continue”.	Open completion screen.

7.3 BUTTON AND STATE RULES

- **Deploy Ingenuity** appears only when Perseverance is stationary and the helicopter is ready to be released.
- **Run Pre-flight Check** appears after deployment.
- **Spin Rotors** appears after the pre-flight check.
- **Take Off** appears after the rotors reach the ready state.
- **Capture Image** appears at the final aerial waypoint.
- **Land** appears when the helicopter is above the marked landing zone.
- Use simple flight controls. A small height / stability indicator is enough; realistic flight physics are not required.

7.4 REQUIRED OBJECTIVES

OBJECTIVE	COMPLETION CONDITION
01 - Deploy and prepare Ingenuity	Player deploys the helicopter and completes the pre-flight check.
02 - Complete the aerial route	Player takes off and reaches the required waypoints.
03 - Capture an image and land	Player captures the aerial image and lands inside the marked zone.

7.5 COMPLETION SCREEN

Title: MISSION COMPLETED - FLIGHT DEMONSTRATION COMPLETE
Learning card: “Ingenuity proved that powered, controlled flight is possible on another planet. It completed 72 flights before rotor damage ended its flying mission in January 2024.”
Buttons: “Replay Mission” and “Return to Mission Selection”.

ACCURACY NOTE

This is a historical recreation, not a live flight simulator. Ingenuity is not currently flying. It traveled to Mars with Perseverance; do not show it launching from Earth as a separate spacecraft. Do not imply that Ingenuity collected rock samples or performed the rover’s science work.

8. SHARED MISSION COMPLETION SCREEN

Use one reusable completion layout across all seven missions.

The completion screen should acknowledge the player's work and reinforce one real scientific idea. Keep it simple and consistent so developers only need to build it once.

ELEMENT	REQUIRED CONTENT / BEHAVIOR
Status title	“MISSION COMPLETED!” plus the mission-specific completion title.
Objective summary	Show the required objectives as completed. Optional bonus tasks appear separately.
Mission fact	Show one or two short, fact-checked points tied to the mission.
Truth label	Use “MISSION FACT” for verified history and “SIMPLIFIED SIMULATION” for game-only representations.
Actions	“Replay Mission” and “Return to Mission Selection”. The return action must always be available.
No forced sharing	Do not require the player to share, sign in, or provide personal details to complete a mission.

8.1 FAILURE, RETRY, AND EXIT RULES

- If a rover moves away from a target, the player may drive back. Do not reset all objectives.
- If an action is performed incorrectly, show one short hint and allow another attempt.
- If the helicopter misses a waypoint, allow the player to turn back and retry the route.
- Pause / Exit must offer a clear confirmation before returning to the selector.
- Restart Mission resets only the currently selected mission.

8.2 COMMON COPY

STATE	SUGGESTED ON-SCREEN TEXT
Before start	“Ready to begin? Your mission has 3 objectives.”
Target not reached	“Move closer to the marked location to continue.”
Action unlocked	“Objective reached. Choose the highlighted action.”
Objective complete	“Objective complete!”
Mission completed	“Mission completed. Explore what this mission helped scientists learn.”
Return prompt	“Return to Mission Selection?”

9. DEVELOPMENT PRIORITIES, ASSETS, AND QA

9.1 BUILD PRIORITY

PRIORITY	BUILD ITEMS
MUST - First playable version	Seven mission cards; briefing screen; correct arrival state; mission-specific required objectives; contextual controls; completion screen; return to selector; short fact card for each mission.
SHOULD - Improve learning	Small mission map; photo capture panels; component / instrument callouts; objective progress feedback; simple waveform for Apollo Seismometer; sample tube / analysis animations.
COULD - Extra polish	Optional bonus targets; badge collection; extra photo gallery; sound effects with mute control; replayable fact cards; extra rover selection details.

9.2 ASSET REQUIREMENTS

- Mission cards: one representative image / model preview for each of the seven missions.
- Apollo Laser Reflectors: lunar module / astronaut, reflector array, Earth-observatory laser inset.
- Apollo Seismometers: astronaut, seismometer, central station, antenna, waveform panel.
- Viking: lander, robotic sampler arm, soil scoop, instrument inlet, panorama / experiment panel.
- Spirit & Opportunity: selectable rover models or a shared twin-rover model, target rocks, Pancam-style photo frames, RAT close-up.
- Curiosity: rover model, robotic arm, drill, sample powder transfer, onboard analysis panel.
- Perseverance: rover model, drill / coring bit, sample tube, tube-sealing animation, cache location marker.
- Ingenuity: helicopter model, deployment scene beneath Perseverance, flight path markers, aerial image frame, landing zone.

9.3 USER-PROVIDED MODEL REFERENCES

Curiosity 3D animation reference: YouTube reference video
Perseverance 3D animation reference: YouTube reference video
These are visual references only. Confirm ownership, permissions, and credit before reusing any footage or model.

9.4 QA CHECKLIST

- Each of the seven cards opens the correct mission and briefing.
- Each mission shows the correct destination and arrival type.
- Viking does not show rover driving after landing.
- Apollo laser source is shown on Earth; the lunar panel only reflects the pulse.
- Apollo seismic waveform is sourced or labeled illustrative.
- Spirit / Opportunity use rock abrasion, not Curiosity / Perseverance-style core collection.
- Curiosity transfers powdered sample to onboard analysis; no sample-cache depot is shown.
- Perseverance collects and seals a core sample tube before the cache objective.
- Ingenuity is deployed from Perseverance and uses flight controls; it is not shown launching from Earth.
- All mission-specific action buttons appear only after their trigger conditions are met.
- Every mission can be paused, restarted, completed, and exited.
- All scientific claims and sample-result text have a source and review date.
- All screen text remains readable on mobile and desktop.

10. FACT-CHECK REFERENCES

Use these sources when writing the final on-screen mission facts. Re-check all facts before the public demo.

1. Apollo laser reflectors - International Laser Ranging Service (ILRS), Apollo and Luna reflector information

https://ilrs.cddis.eosdis.nasa.gov/missions/satellite_missions/current_missions/ap11_reflector.html

Reflector array locations, corner-cube construction, and the Apollo 11 / 14 / 15 arrays.

2. Apollo laser ranging - NASA / U.S. Government Lunar Historic Sites report

https://www.nasa.gov/wp-content/uploads/2023/12/nasa-usg-lunar-historic-sites-reva.pdf

Confirms lunar laser retroreflector ranging is an ongoing experiment and identifies Apollo sites.

3. Apollo 11 Passive Seismic Experiment - NASA Planetary Data System

https://pds.nasa.gov/ds-view/pds/viewContext.jsp?identifier=urn%3Anasa%3Apds%3Acontext%3Ainstrument%3Apse.a11e&version=1.0

Apollo 11 early seismometer and detection of lunar seismic activity.

4. Apollo 12 Passive Seismic Experiment - NASA Planetary Data System

https://pds.nasa.gov/ds-view/pds/viewContext.jsp?identifier=urn%3Anasa%3Apds%3Acontext%3Ainstrument%3Apse.a12a&version=1.0

ALSEP passive seismic station details.

5. Apollo passive seismic data - DARTS / JAXA

https://darts.isas.jaxa.jp/en/datasets/apollo-pse-data

Archived Apollo seismic data coverage for the Apollo lunar seismic experiments.

6. Viking Project - NASA Science

https://science.nasa.gov/mission/viking/

Viking mission purpose, lander activities, experiments, and limits of life-detection results.

7. Viking Spacecraft and Science - NASA Science

https://science.nasa.gov/mission/viking/spacecraft-and-science/

Lander images, soil sampling, experiments, meteorology, and seismometer deployment.

8. Viking 1 - NASA Science

https://science.nasa.gov/mission/viking-1/

Viking 1 robotic arm sampling and mission communications history.

9. Spirit and Opportunity Science Instruments - NASA Science

https://science.nasa.gov/mission/mars-exploration-rovers-spirit-and-opportunity/science-instruments/

Pancam, microscopic imager, and Rock Abrasion Tool functions.

10. Spirit - NASA Science

https://science.nasa.gov/mission/mer-spirit/

Spirit mission objective, instruments, and mission history.

11. Curiosity Science Instruments - NASA Science

https://science.nasa.gov/mission/msl-curiosity/science-instruments/

Curiosity cameras, instruments, and science capabilities.

12. Curiosity drill samples - NASA Science

https://science.nasa.gov/resource/curiositys-42-drill-holes/

Drilled powder samples and transfer to CheMin / SAM instruments.

13. Curiosity drill system - NASA / JPL

https://mars.nasa.gov/multimedia/interactives/learncuriosity/index-2.html

How Curiosity's drill powders rock for onboard analysis.

14. Perseverance Rover Components - NASA Science

https://science.nasa.gov/mission/mars-2020-perseverance/rover-components/

Drill, sample handling, sample tubes, sealing, and caching process.

15. Perseverance Rover - NASA Science

https://science.nasa.gov/mission/mars-sample-return/perseverance-rover/

Mission goal to collect and cache Mars samples for possible future retrieval.

16. Ingenuity Mars Helicopter - NASA Science

https://science.nasa.gov/mission/mars-2020-perseverance/ingenuity-mars-helicopter/

Ingenuity's flight objective, 72 flights, and mission timeline.

17. Ingenuity mission ended - NASA

https://www.nasa.gov/news-release/after-three-years-on-mars-nasas-ingenuity-helicopter-mission-ends/

Final flight, rotor damage, and end of mission announcement.

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END OF DEVELOPER BRIEF

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